

Project Design Phase

Solution Architecture

Project Name: Import Data Using Transform Maps (Spreadsheet)

Solution Architecture:

Solution architecture is a complex process – with many sub-processes – that bridges the gap between business problems and technology solutions. Its goals are to:

- **Find the best tech solution to solve existing business problems:** Automate and streamline bulk data ingestion by uploading external spreadsheet records (Excel/CSV) directly into target database tables, eliminating human error and manual data entry.
- **Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders:** Establish a clear data ingestion pipeline starting from raw file upload through staging tables, transformation rules, and final target record population.
- **Define features, development phases, and solution requirements:** Configure field-to-field mapping rules, identify Coalesce unique identifiers to prevent record duplication, and set up transformation logic for data consistency.
- **Provide specifications according to which the solution is defined, managed, and delivered:** Detail logging mechanisms, staging table cleanup, error handling, and security role access required for execution.

Example - Solution Architecture Diagram:

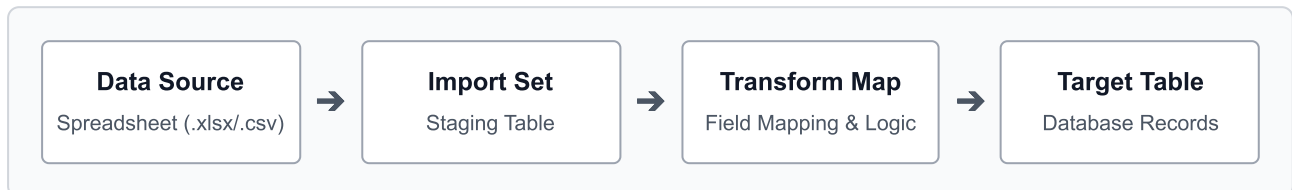


Figure 1: Architecture and data flow of the Import Data Using Transform Maps (Spreadsheet) application